# Experiment 1: point forecasts of wind with complex ARIMA

Wind Forecasting with Complex ARIMA — working notes

30 September 2026 (runs 1–4)

All numbers below come from the scripts in `R/` and the tables in `results/tables/`; the commands to reproduce them are in Section 9.

## 1 Aim

The paper argues that wind should be modelled as *one complex variable* (speed and direction together) with complex ARIMA (cARIMA), rather than as two real series. Experiment 1 is the first test of this: rolling-origin point forecasts of 10-minute wind, from 10 minutes to 6 hours ahead, comparing cARIMA with persistence and with pairs of independent real ARIMA models.

## 2 Data

- Kaggle Wind Turbine SCADA dataset, file `T1.csv`: one turbine in Turkey, 2018, 10-minute resolution, wind speed (m/s) and wind direction (degrees). Details are in `data/README.md`.
- 50,530 rows on a 52,560-slot grid: 2,030 missing slots, including three gaps longer than 3 days.
- `R/02_prepare_data.R`:
  - puts the data on a regular 10-minute grid;
  - fills gaps of at most 36 slots (6 hours) by linear interpolation of $\operatorname{Re} Z_t$ and $\operatorname{Im} Z_t$ (defined below);
  - replaces speeds below 0.1 m/s (10 exact zeros) by 0.1, so that logarithms exist;
  - treats an origin as *eligible* if its 4-week training window has no missing values, and its 6-hour horizon has no missing values and no values from interpolations longer than 3 slots.
- The first eligible window (2018-01-30 14:40 to 2018-02-27 14:30) is the *selection window*. The 300 test origins are spread evenly over the later eligible origins (2018-02-27 14:40 to 2018-12-31 17:50; none in October, because of gaps).

## 3 How the wind is modelled

### 3.1 Wind as a complex number

With speed $v_t > 0$ and direction angle $\theta_t \in (-\pi, \pi]$, the wind is

$$Z_t = v_t e^{\mathrm{i}\theta_t} = v_t \cos\theta_t + \mathrm{i}\, v_t \sin\theta_t. \tag{1}$$

$\operatorname{Re} Z_t$ and $\operatorname{Im} Z_t$ are the plain Cartesian coordinates of the wind vector (not the meteorological zonal and meridional components).

### 3.2 Complex ARIMA

The cARIMA$(p, d, q)$ model is the ARIMA model with the variable and all parameters complex:

$$\Big(1 - \sum_{j=1}^{p} \phi_j B^j\Big)(1 - B)^d \, y_t = \beta_0 + \Big(1 + \sum_{j=1}^{q} \theta_j B^j\Big)\varepsilon_t, \qquad \phi_j, \theta_j, \beta_0 \in \mathbb{C}, \tag{2}$$

where $B$ is the backshift operator. It is estimated by maximum likelihood with `complex::clm()`; in run 4 the orders are selected with `complex::auto.carima()`.

Multiplying by a complex coefficient $\phi = a + \mathrm{i}b = re^{\mathrm{i}\alpha}$ *scales* by $r$ and *rotates* by $\alpha$. Written for the pair $(\operatorname{Re} y, \operatorname{Im} y)$,

$$\begin{pmatrix} \operatorname{Re}(\phi y) \\ \operatorname{Im}(\phi y) \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} \operatorname{Re} y \\ \operatorname{Im} y \end{pmatrix}, \tag{3}$$

so cARIMA is a restricted bivariate VARIMA with 2 real parameters per lag instead of 4. This parsimony relative to VAR is the central argument of the paper.

The error $\varepsilon_t$ is complex normal with variance $\sigma^2 = \mathrm{E}\,|\varepsilon_t|^2$ and pseudo-variance $\tau = \mathrm{E}\,\varepsilon_t^2$ (for zero-mean errors). In terms of the real and imaginary parts,

$$\operatorname{Var}(\operatorname{Re}\varepsilon_t) = \tfrac{1}{2}(\sigma^2 + \operatorname{Re}\tau), \quad \operatorname{Var}(\operatorname{Im}\varepsilon_t) = \tfrac{1}{2}(\sigma^2 - \operatorname{Re}\tau), \quad \operatorname{Cov}(\operatorname{Re}\varepsilon_t, \operatorname{Im}\varepsilon_t) = \tfrac{1}{2}\operatorname{Im}\tau. \tag{4}$$

If $\tau \neq 0$ the error is *non-circular*: its scatter is an ellipse rather than a circle. For wind this is anisotropy, a prevailing direction of change. We summarise it by the index

$$\kappa = \frac{|\tau|}{\sigma^2} \in [0, 1], \tag{5}$$

which is 0 for circular errors and 1 for errors lying on a line.

### 3.3 Why logarithms

The main target is the complex logarithm

$$\log Z_t = \log v_t + \mathrm{i}\,\tilde{\theta}_t, \tag{6}$$

where $\tilde{\theta}_t$ is the unwrapped angle (Section 3.4).

- It *separates speed and direction*: the real part is log speed, the imaginary part is the angle. A cARIMA on $\log Z_t$ is still one complex model: the two parts are linked through the complex coefficients and the error covariance.
- Changes of speed are roughly *multiplicative*: they are larger when it is windy, so log speed is a more natural scale than speed.
- The back-transform $\hat{Z}_{t+h} = \exp(\widehat{\log Z}_{t+h})$ always gives a positive speed and a valid angle.
- $\exp(\widehat{\log Z}_{t+h})$ is not the conditional mean of $Z_{t+h}$. This experiment does not correct for that.

The models on $Z_t$ itself are kept for comparison.

### 3.4 Why differences, and angle unwrapping

- 10-minute wind is very persistent: over a 4-week window the level wanders without a stable mean, and persistence is a hard benchmark. So in runs 1–3 all models use $d = 1$. With logarithms the differences have a clear meaning:

$$(1 - B)\log Z_t = \log \frac{v_t}{v_{t-1}} + \mathrm{i}\,(\tilde{\theta}_t - \tilde{\theta}_{t-1}), \tag{7}$$

  the log ratio of speeds and the change of direction.

- The angle in $(-\pi, \pi]$ jumps by about $2\pi$ whenever the wind crosses the cut (696 times in `T1.csv`). In the differences these would be huge artificial shocks. Therefore, within each training window $t = 1, \ldots, T$, the angle is unwrapped:

$$\tilde{\theta}_1 = \theta_1, \qquad \tilde{\theta}_t = \tilde{\theta}_{t-1} + \operatorname{wrap}(\theta_t - \theta_{t-1}), \qquad \operatorname{wrap}(x) = \operatorname{atan2}(\sin x, \cos x), \tag{8}$$

  so that $\tilde{\theta}_t - \tilde{\theta}_{t-1}$ is the shortest-arc change. Forecasts are wrapped back to $(-\pi, \pi]$ for evaluation.

- `clm()` estimates the model in the "direct" form: forecasts are produced in levels, and models with different $d$ are comparable by information criteria.

- In run 4, `auto.carima()` may choose $d \in \{0, 1\}$. It picks $d = 0$ for $Z_t$, with $\phi_1 \approx 0.993$–$0.994$. The half-life of a deviation from the mean, $\log 0.5 / \log |\phi_1|$, is about 100–120 steps, i.e. 16–20 hours. For $\log Z_t$ it picks $d = 1$.

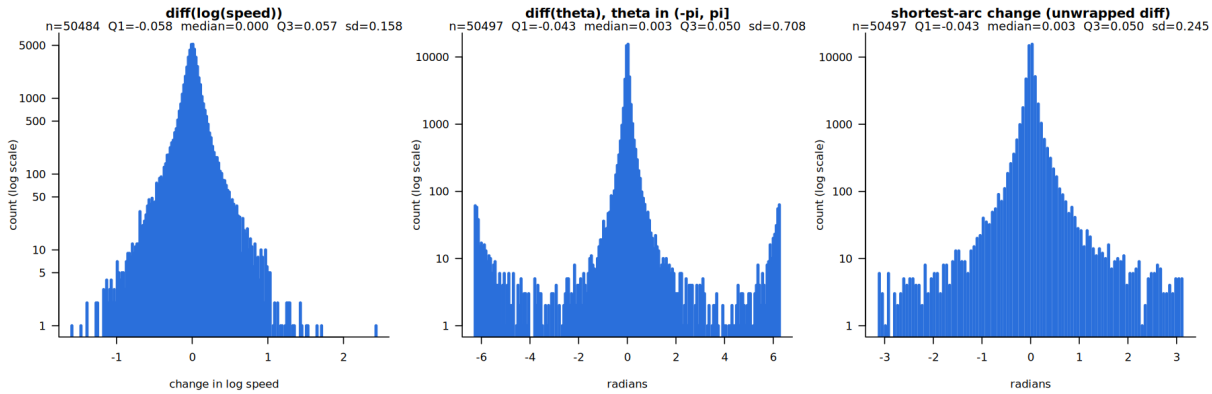



**Figure 1:** 10-minute changes in `T1.csv` (log-scale counts): log speed; the wrapped angle difference, with its $\pm 2\pi$ jumps; and the shortest-arc angle change.

### 3.5 Why the complex generalised normal distribution

The 10-minute changes have very heavy tails. Table 1 gives the kurtosis $\mathrm{E}(x - \mathrm{E}\,x)^4 / (\mathrm{Var}\,x)^2$ of consecutive changes in `T1.csv`; it is 3 for a normal distribution. About 3% of angle changes exceed 30° and 0.5% exceed 90°.

**Table 1:** Kurtosis of 10-minute changes (normal: 3).

| change | kurtosis |
|---|---|
| $\log v_t - \log v_{t-1}$ | 14.4 |
| $\operatorname{wrap}(\theta_t - \theta_{t-1})$ | 52.8 |
| $\mathrm{Re}\, Z_t - \mathrm{Re}\, Z_{t-1}$ | 131.9 |
| $\mathrm{Im}\, Z_t - \mathrm{Im}\, Z_{t-1}$ | 25.5 |

Under a normal likelihood these outliers pull the estimates, in particular the constant. For $\log Z_t$ with $d = 1$, $\mathrm{Im}\,\beta_0$ is a steady rotation of the forecasts: 0.0122 rad (0.70°) per step in `carima_logZ`, i.e. about 25° over the 6-hour horizon. (The project notes for run 2 record a median rotation of 17° at $h = 36$ for the normal model and 5° with the generalised normal; this was not recomputed here.)

We therefore also use the complex generalised normal distribution (`distribution = "dcgnorm"` in `clm()`; Novey, Adali and Roy, 2010). For the error $e = z - \mu$, with scale $\alpha^2$,

pseudo-scale $\psi^2$ and shape $\beta$,

$$f(z) = \frac{2\beta}{\pi\,\Gamma(1/\beta)\,2^{1/\beta}\sqrt{\alpha^4 - |\psi^2|^2}}\exp\!\Big(-\tfrac{1}{2}Q^{\beta}\Big), \qquad Q = \frac{2\big(\alpha^2|e|^2 - \mathrm{Re}(\overline{\psi^2}\,e^2)\big)}{\alpha^4 - |\psi^2|^2}. \tag{9}$$

For $\beta = 1$ this is the complex normal distribution with variance $\alpha^2$ and pseudo-variance $\psi^2$; $\beta < 1$ gives heavier tails ($\beta = 0.5$ is Laplace-like). The variance and pseudo-variance are $\alpha^2$ and $\psi^2$ multiplied by $2^{1/\beta}\Gamma(2/\beta)/(2\Gamma(1/\beta))$. The shape is estimated, while the scale and pseudo-scale are concentrated out in closed form. The estimates are $\hat\beta \approx 0.21$ for $\log Z_t$ and $\hat\beta \approx 0.41$ for $Z_t$, far from normal. On the selection window the generalised normal raises the log-likelihood by about 5,000 ($\log Z_t$) and 1,500 ($Z_t$).

# 4 Experimental design

The pipeline is `R/02_prepare_data.R` → `R/03_select_orders.R` → `R/04_rolling_origin.R` → `R/05_evaluate.R`. Shared code is in `R/functions.R`, and the parameter tables come from `R/06_parameters.R`.

- **Rolling origin:** 300 origins, a window of 4 weeks ($T = 4032$ observations), horizons $h = 1, \ldots, 36$ (10 minutes to 6 hours). Parameters are re-estimated at every origin.
- **Order selection** is done once, on the selection window; the orders are then fixed for all origins.
  - cARIMA, runs 1–3: $d = 1$, and $p, q \in \{0, \ldots, 3\}$ by AICc. From its default start, `clm()` can end in a poor local optimum, so each fit also starts from the estimates of the nested models $(p-1, q)$ and $(p, q-1)$ and keeps the highest log-likelihood.
  - cARIMA, run 4: stepwise search with

    ```
    auto.carima(y, orders = list(ar = 3, i = 1, ma = 3), constant = NULL),
    ```

    which chooses $p, q \le 3$, $d \le 1$ and whether to include $\beta_0$.
  - Real benchmarks: `forecast::auto.arima` with $d = 1$; `smooth::auto.msarima` on the differenced series, then refitted with $d = 1$.
  - At each origin, cARIMA fits start from the default and from the selection-window estimates.
- **Error measures**, in original units, for origin $t$ and horizon $h$:

$$e^{\text{speed}}_{t,h} = |\hat Z_{t+h|t}| - |Z_{t+h}| \quad \text{(m/s)}, \tag{10}$$

$$e^{\text{angle}}_{t,h} = \tfrac{180}{\pi}\,\mathrm{wrap}(\mathrm{Arg}\,\hat Z_{t+h|t} - \mathrm{Arg}\,Z_{t+h}) \quad \text{(degrees)}, \tag{11}$$

$$e^{\text{vector}}_{t,h} = |\hat Z_{t+h|t} - Z_{t+h}| \quad \text{(m/s)}, \tag{12}$$

  and $\mathrm{RMSE}_h = \big(\frac{1}{300}\sum_t e^2_{t,h}\big)^{1/2}$. The column "h1–36" pools all horizons.

**Table 2:** Models. The real benchmarks forecast two series independently and combine them into $\hat{Z}$: (speed, angle), (log speed, angle) and $(\mathrm{Re}\, Z, \mathrm{Im}\, Z)$. Run 1: cARIMA with a constant and the real benchmarks; run 2: generalised normal; run 3: $\log Z$ without a constant (`_nd`); run 4: `auto.carima`.

| model | series | distribution | orders | constant $\beta_0$ |
|---|---|---|---|---|
| `persistence` | – | – | $\hat{Z}_{t+h\|t} = Z_t$ | – |
| `carima_logZ` | $\log Z$ | normal | $d = 1$, AICc | yes |
| `carima_logZ_cgn` | $\log Z$ | generalised normal | $d = 1$, AICc | yes |
| `carima_logZ_nd` | $\log Z$ | normal | $d = 1$, AICc | no |
| `carima_logZ_cgn_nd` | $\log Z$ | generalised normal | $d = 1$, AICc | no |
| `carima_z` | $Z$ | normal | $d = 1$, AICc | yes |
| `carima_z_cgn` | $Z$ | generalised normal | $d = 1$, AICc | yes |
| `auto_carima_logZ` | $\log Z$ | normal | `auto.carima` | selected |
| `auto_carima_logZ_cgn` | $\log Z$ | generalised normal | `auto.carima` | selected |
| `auto_carima_z` | $Z$ | normal | `auto.carima` | selected |
| `auto_carima_z_cgn` | $Z$ | generalised normal | `auto.carima` | selected |
| `arima_*`, `msarima_*` | two real series | normal | `auto.arima` / `auto.msarima`, $d = 1$ | selected |

# 5 Selected orders and estimated parameters

Table 3 gives the estimates on the selection window (`results/tables/parameters_carima.csv`); in the rolling origin they are re-estimated at every origin. The coefficients follow the signs in (2). Standard errors are not reported: `summary()` fails on these fits in the current package version (its `vcov` refit cannot find `distribution`).

**Table 3:** cARIMA estimates on the selection window. $\hat{\beta}$ is the shape of the generalised normal.

| model | $(p, d, q)$ | $\hat{\beta}_0$ | $\hat{\phi}_1$ | $\hat{\theta}_1, \ldots, \hat{\theta}_q$ | $\hat{\beta}$ | log-likelihood |
|---|---|---|---|---|---|---|
| `carima_logZ` | (0,1,3) | $-0.0003 + 0.0122$i | – | $-0.0856 - 0.0148$i<br>$-0.0511 - 0.0018$i<br>$-0.0250 + 0.0011$i | – | 493.33 |
| `carima_logZ_cgn` | (0,1,2) | $0.0001 + 0.0012$i | – | $0.0607 + 0.0056$i<br>$-0.0649 + 0.0047$i | 0.212 | 5510.52 |
| `carima_z` | (0,1,2) | $-0.0026 - 0.0005$i | – | $-0.0281 - 0.0245$i<br>$-0.1250 + 0.0375$i | – | −9906.10 |
| `carima_z_cgn` | (0,1,2) | $-0.0016 + 0.0088$i | – | $0.0306 + 0.0023$i<br>$-0.1000 + 0.0250$i | 0.408 | −8420.80 |
| `carima_logZ_nd` | (1,1,2) | none | $0.6264 - 0.0111$i | $-0.7074 + 0.0000$i<br>$0.0000 + 0.0000$i | – | 492.51 |
| `carima_logZ_cgn_nd` | (0,1,2) | none | – | $0.0650 + 0.0089$i<br>$-0.0774 + 0.0131$i | 0.212 | 5509.75 |
| `auto_carima_z` | (1,0,2) | $-0.0159 + 0.0138$i | $0.9930 + 0.0022$i | $-0.0270 - 0.0287$i<br>$-0.1293 + 0.0394$i | – | −9888.61 |
| `auto_carima_z_cgn` | (1,0,2) | $-0.0102 + 0.0212$i | $0.9943 + 0.0006$i | $0.0344 + 0.0010$i<br>$-0.0918 + 0.0197$i | 0.409 | −8408.73 |
| `auto_carima_logZ` | (1,1,1) | $0.0000 + 0.0044$i | $0.6427 + 0.0044$i | $-0.7250 - 0.0145$i | – | 499.72 |
| `auto_carima_logZ_cgn` | (0,1,2) | none | – | $0.0685 + 0.0017$i<br>$-0.0710 + 0.0003$i | 0.213 | 5509.95 |

- **Constants.** For $\log Z_t$, $\mathrm{Im}\, \hat{\beta}_0$ is the angle drift per 10 minutes: 0.0122 rad for `carima_logZ`, the same as the drift of the real `arima_angle` model (0.0121, Table 5). The generalised normal shrinks it to 0.0012. `auto.carima` keeps the constant for $\log Z_t$ with the normal distribution and drops it with the generalised normal.
- $Z_t$ **with** $d = 0$**.** $\hat{\phi}_1$ is close to 1 with an almost zero argument: close to a unit root, without rotation. The implied long-run mean $\mu = \beta_0/(1 - \phi_1)$ is $-2.63 + 1.14$i (normal) and $-2.16 + 3.49$i (generalised normal), vectors of 2.9 and 4.1 m/s. For comparison, the

mean *speed* in the window is 8.5 m/s, but the mean *vector* is only 3.5 m/s long, because winds from different directions cancel out. Reverting to the mean therefore shrinks $|\hat{Z}|$ at long horizons.

- `auto_carima_logZ_cgn` selected $(0, 1, 2)$ without a constant, the same specification as `carima_logZ_cgn_nd`.
- **Caveat, `carima_logZ_nd`.** Refitting $(1, 1, 2)$ reaches a log-likelihood of 492.51, above the 489.91 found during order selection (`carima_order_grid.csv`). The best fit has $\hat{\theta}_2 = 0$ exactly, which suggests the optimiser stayed at the starting values padded from the nested $(1, 1, 1)$ model. So the order selection for this model used a sub-optimal fit, and $(1, 1, 1)$ may have been the better choice. All other models reproduce the selection log-likelihood to within 0.02.

Table 4 gives the sample covariance of the residuals $(\operatorname{Re}\hat{\varepsilon}_t, \operatorname{Im}\hat{\varepsilon}_t)$ and the non-circularity index (5), computed from the centred residuals $\tilde{\varepsilon}_t = \hat{\varepsilon}_t - \frac{1}{T}\sum_s \hat{\varepsilon}_s$ as $\hat{\kappa} = \left|\sum_t \tilde{\varepsilon}_t^2\right| / \sum_t |\tilde{\varepsilon}_t|^2$. Values of 0.27–0.33 mean clearly elliptical errors, i.e. a prevailing direction of change, which a pair of independent real models ignores.

**Table 4:** Residuals on the selection window: standard deviations and correlation of the real and imaginary parts, and the non-circularity $\kappa$.

| model | sd $\operatorname{Re}\hat{\varepsilon}$ | sd $\operatorname{Im}\hat{\varepsilon}$ | cor$(\operatorname{Re}\hat{\varepsilon}, \operatorname{Im}\hat{\varepsilon})$ | $\kappa$ |
|---|---|---|---|---|
| `carima_logZ` | 0.196 | 0.265 | −0.076 | 0.304 |
| `carima_logZ_cgn` | 0.201 | 0.264 | −0.054 | 0.271 |
| `carima_z` | 0.901 | 0.798 | 0.310 | 0.330 |
| `carima_z_cgn` | 0.904 | 0.800 | 0.311 | 0.332 |
| `carima_logZ_nd` | 0.195 | 0.265 | −0.075 | 0.306 |
| `carima_logZ_cgn_nd` | 0.201 | 0.264 | −0.051 | 0.269 |
| `auto_carima_z` | 0.900 | 0.796 | 0.310 | 0.331 |
| `auto_carima_z_cgn` | 0.903 | 0.799 | 0.313 | 0.334 |
| `auto_carima_logZ` | 0.195 | 0.266 | −0.075 | 0.306 |
| `auto_carima_logZ_cgn` | 0.201 | 0.264 | −0.053 | 0.269 |

**Table 5:** Real benchmarks on the selection window (`results/tables/parameters_benchmarks.csv`). $\phi_j$: AR, $\theta_j$: MA (sign convention $1 + \theta_1 B + \dots$), $c$: drift.

| model | $(p, d, q)$ | drift | coefficients |
|---|---|---|---|
| `arima_speed` | (5,1,3) | no | $\phi_1 = -0.5156$, $\phi_2 = 0.1343$, $\phi_3 = 0.7397$, $\phi_4 = 0.0330$, $\phi_5 = 0.0443$, $\theta_1 = 0.4364$, $\theta_2 = -0.2834$, $\theta_3 = -0.7965$ |
| `msarima_speed` | (3,1,2) | yes | $\phi_1 = 0.3978$, $\phi_2 = 0.2726$, $\phi_3 = 0.0589$, $\theta_1 = -0.4786$, $\theta_2 = -0.3482$, $c = -0.0002$ |
| `arima_logSpeed` | (1,1,2) | no | $\phi_1 = 0.7313$, $\theta_1 = -0.9495$, $\theta_2 = 0.0767$ |
| `msarima_logSpeed` | (3,1,3) | yes | $\phi_1 = 0.2165$, $\phi_2 = 0.1815$, $\phi_3 = 0.1305$, $\theta_1 = -0.4353$, $\theta_2 = -0.2192$, $\theta_3 = -0.1209$, $c = -0.0001$ |
| `arima_angle` | (2,1,0) | yes | $\phi_1 = 0.0346$, $\phi_2 = -0.0267$, $c = 0.0121$ |
| `msarima_angle` | (0,1,1) | yes | $\theta_1 = 0.0355$, $c = 0.0121$ |
| `arima_re` | (2,1,1) | no | $\phi_1 = -0.3312$, $\phi_2 = -0.1251$, $\theta_1 = 0.2956$ |
| `msarima_re` | (0,1,2) | yes | $\theta_1 = -0.0360$, $\theta_2 = -0.1247$, $c = -0.0022$ |
| `arima_im` | (2,1,2) | no | $\phi_1 = 0.1092$, $\phi_2 = -0.0453$, $\theta_1 = -0.1265$, $\theta_2 = -0.1186$ |
| `msarima_im` | (0,1,2) | yes | $\theta_1 = -0.0196$, $\theta_2 = -0.1664$, $c = -0.0004$ |

# 6 Results

Tables 6–8 give the RMSE over the 300 origins at selected horizons. All 36 horizons are in `results/tables/rmse_speed.csv`, `rmse_angle.csv` and `rmse_z.csv`. Bold marks the best model in each column. No model failed at any origin.

**Table 6:** RMSE of speed $|\hat{Z}|$ (m/s).

| model | $h = 1$ | $h = 3$ | $h = 6$ | $h = 12$ | $h = 24$ | $h = 36$ | $h = 1$–$36$ |
|---|---|---|---|---|---|---|---|
| `persistence` | 0.710 | 1.048 | 1.571 | 1.995 | 2.607 | 3.082 | 2.282 |
| `carima_logZ` | 0.715 | 1.061 | 1.575 | 2.000 | 2.602 | 3.085 | 2.283 |
| `carima_logZ_cgn` | **0.706** | 1.047 | 1.575 | 2.014 | 2.634 | 3.129 | 2.309 |
| `carima_logZ_nd` | 0.714 | 1.058 | 1.576 | 1.996 | 2.593 | 3.071 | 2.276 |
| `carima_logZ_cgn_nd` | 0.706 | **1.045** | 1.569 | 1.991 | 2.609 | 3.083 | 2.284 |
| `carima_z` | 0.718 | 1.055 | 1.569 | 1.989 | 2.598 | 3.080 | 2.281 |
| `carima_z_cgn` | 0.716 | 1.054 | 1.574 | 1.999 | 2.613 | 3.093 | 2.294 |
| `auto_carima_logZ` | 0.715 | 1.067 | 1.582 | 2.013 | 2.637 | 3.178 | 2.321 |
| `auto_carima_logZ_cgn` | 0.707 | 1.047 | 1.570 | 1.992 | 2.608 | 3.083 | 2.284 |
| `auto_carima_z` | 0.726 | 1.083 | 1.615 | 2.105 | 2.818 | 3.354 | 2.466 |
| `auto_carima_z_cgn` | 0.720 | 1.062 | 1.584 | 2.031 | 2.690 | 3.183 | 2.352 |
| `arima_logSpeed_angle` | 0.727 | 1.083 | 1.598 | 2.014 | 2.586 | 3.023 | 2.271 |
| `arima_re_im` | 0.727 | 1.054 | 1.569 | 1.978 | 2.576 | 3.044 | 2.256 |
| `arima_speed_angle` | 0.720 | 1.047 | **1.557** | **1.951** | **2.500** | **2.981** | **2.214** |
| `msarima_logSpeed_angle` | 0.728 | 1.087 | 1.607 | 2.022 | 2.596 | 3.049 | 2.283 |
| `msarima_re_im` | 0.728 | 1.061 | 1.574 | 1.996 | 2.603 | 3.085 | 2.283 |
| `msarima_speed_angle` | 0.717 | 1.051 | 1.567 | 1.976 | 2.557 | 3.033 | 2.247 |

**Table 7:** RMSE of angle (degrees).

| model | $h = 1$ | $h = 3$ | $h = 6$ | $h = 12$ | $h = 24$ | $h = 36$ | $h = 1$–$36$ |
|---|---|---|---|---|---|---|---|
| `persistence` | 12.42 | 20.74 | 26.78 | 36.49 | 43.01 | 46.14 | 38.59 |
| `carima_logZ` | 12.67 | 20.71 | 26.73 | 36.99 | 45.42 | 49.47 | 40.18 |
| `carima_logZ_cgn` | 12.50 | 20.88 | 26.82 | 36.57 | 43.41 | 46.31 | 38.74 |
| `carima_logZ_nd` | 12.64 | **20.44** | **26.29** | **36.14** | 42.63 | **45.77** | **38.21** |
| `carima_logZ_cgn_nd` | 12.46 | 20.90 | 26.84 | 36.60 | 43.02 | 46.13 | 38.62 |
| `carima_z` | 12.81 | 20.96 | 26.82 | 36.53 | 42.82 | 45.94 | 38.50 |
| `carima_z_cgn` | 12.66 | 21.00 | 26.96 | 36.27 | **42.57** | 46.10 | 38.40 |
| `auto_carima_logZ` | 12.52 | 20.83 | 27.04 | 37.24 | 45.58 | 49.57 | 40.29 |
| `auto_carima_logZ_cgn` | 12.48 | 20.87 | 26.82 | 36.55 | 42.99 | 46.12 | 38.59 |
| `auto_carima_z` | 12.98 | 21.47 | 27.64 | 36.63 | 44.31 | 48.28 | 39.80 |
| `auto_carima_z_cgn` | 12.68 | 21.13 | 27.13 | 36.43 | 43.02 | 46.80 | 38.75 |
| `arima_logSpeed_angle` | 12.49 | 21.09 | 27.15 | 37.27 | 45.64 | 49.42 | 40.38 |
| `arima_re_im` | 12.87 | 21.14 | 27.06 | 36.75 | 42.60 | 45.84 | 38.57 |
| `arima_speed_angle` | 12.49 | 21.09 | 27.15 | 37.27 | 45.64 | 49.42 | 40.38 |
| `msarima_logSpeed_angle` | **12.38** | 21.15 | 27.29 | 37.27 | 45.68 | 49.43 | 40.40 |
| `msarima_re_im` | 12.74 | 20.85 | 26.78 | 36.58 | 42.94 | 45.99 | 38.56 |
| `msarima_speed_angle` | **12.38** | 21.15 | 27.29 | 37.27 | 45.68 | 49.43 | 40.40 |

**Table 8:** RMSE of the wind vector $|\hat{Z} - Z|$ (m/s).

| model | $h = 1$ | $h = 3$ | $h = 6$ | $h = 12$ | $h = 24$ | $h = 36$ | $h = 1$–$36$ |
|---|---|---|---|---|---|---|---|
| `persistence` | 0.960 | 1.634 | 2.307 | 2.996 | 4.104 | 4.721 | 3.567 |
| `carima_logZ` | 0.965 | 1.658 | 2.363 | 3.139 | 4.507 | 5.385 | 3.915 |
| `carima_logZ_cgn` | **0.955** | **1.620** | 2.298 | 2.998 | 4.126 | 4.763 | 3.584 |
| `carima_logZ_nd` | 0.961 | 1.631 | 2.297 | 2.991 | 4.078 | 4.700 | 3.550 |
| `carima_logZ_cgn_nd` | 0.955 | 1.620 | 2.294 | 2.981 | 4.093 | 4.710 | 3.555 |
| `carima_z` | 0.964 | 1.630 | 2.296 | 2.979 | 4.077 | 4.699 | 3.549 |
| `carima_z_cgn` | 0.963 | 1.635 | 2.306 | 2.984 | 4.091 | 4.715 | 3.560 |
| `auto_carima_logZ` | 0.965 | 1.676 | 2.388 | 3.186 | 4.580 | 5.535 | 3.987 |
| `auto_carima_logZ_cgn` | 0.955 | 1.622 | 2.296 | 2.983 | 4.093 | 4.711 | 3.556 |
| `auto_carima_z` | 0.967 | 1.633 | 2.307 | 2.985 | 4.030 | 4.587 | 3.509 |
| `auto_carima_z_cgn` | 0.964 | 1.634 | 2.299 | **2.946** | **3.977** | **4.501** | **3.455** |
| `arima_logSpeed_angle` | 0.971 | 1.684 | 2.390 | 3.150 | 4.488 | 5.360 | 3.909 |
| `arima_re_im` | 0.974 | 1.631 | **2.294** | 2.969 | 4.045 | 4.648 | 3.522 |
| `arima_speed_angle` | 0.968 | 1.669 | 2.374 | 3.132 | 4.468 | 5.360 | 3.900 |
| `msarima_logSpeed_angle` | 0.973 | 1.693 | 2.406 | 3.162 | 4.500 | 5.380 | 3.923 |
| `msarima_re_im` | 0.973 | 1.645 | 2.309 | 2.995 | 4.098 | 4.723 | 3.567 |
| `msarima_speed_angle` | 0.965 | 1.673 | 2.378 | 3.132 | 4.480 | 5.365 | 3.901 |

# 7 Findings

1. **No model beats persistence on all three measures.** Gains over persistence are small, at most about 3% at $h = 1$–$36$ (vector: 3.455 vs 3.567), as expected for 10-minute wind with no exogenous information. Badly specified models lose up to about 10%.

2. **Short horizons ($h = 1$–$3$):** cARIMA on $\log Z$ with the generalised normal (`carima_logZ_cgn`, `carima_logZ_cgn_nd`) is best for the vector error and speed.

3. **Drift rotates forecasts.** With a normal likelihood, a constant for $\log Z$ is a steady rotation of the direction. `carima_logZ` and `auto_carima_logZ` are among the worst for the angle (about 40° vs 38.6° for persistence at $h = 1$–$36$) and for the vector error. Without the constant, `carima_logZ_nd` has the best angle of all models (38.21°). The generalised normal largely removes the problem even with a constant.

4. **$Z$ with $d = 0$ (run 4):** `auto_carima_z_cgn` has the best vector error of all models at $h = 1$–$36$ (3.455, vs 3.567 for persistence and 3.522 for the best real pair, `arima_re_im`), and the advantage comes from $h = 12$ onwards. It pays in speed (2.352 vs 2.282), because it reverts to the short mean wind vector. `auto_carima_z` (normal) does the same with a worse speed (2.466, the worst).

5. **Speed alone** is best forecast by the real `arima_speed_angle` from $h = 6$ onwards (2.214 at $h = 1$–$36$); it models speed directly.

6. **Heavy tails are real:** $\hat{\beta} \approx 0.2$ ($\log Z$) and 0.4 ($Z$), and the generalised normal helps most where the drift of the normal model was hurting.

## 8 Limitations and next steps

- Point forecasts only: no predictive distributions, no statistical tests (e.g. Diebold–Mariano), no MAE or relative measures yet.
- One turbine and one year; orders are selected on a single February window and fixed for the year.
- The exp(·) back-transform of the log $Z$ forecasts is not bias-corrected.
- The order selection for `carima_logZ_nd` used a sub-optimal fit (Section 5).
- Not yet: diurnal and Fourier terms, VAR and VARMA benchmarks (the parsimony argument needs them), power forecasts, other datasets.

## 9 Reproduce

```
Rscript R/02_prepare_data.R              # needs data/raw/T1.csv
Rscript R/03_select_orders.R [model ...] # no arguments = all models
Rscript R/04_rolling_origin.R [model ...]
Rscript R/05_evaluate.R
Rscript R/06_parameters.R
```

`data/README.md` explains how to download `T1.csv`. Scripts `03` and `04` merge their results into the existing ones by model name. Run 4 (four `auto.carima` models) took 23 minutes for order selection and 2.8 hours for the rolling origin on 4 cores.